# Notes: Atkin, Khandelwal & Osman (QJE, 2017)

## Exporting and Firm Performance: Evidence from a Randomized Experiment

## Contents


**1 Big picture** **3**

1.1 Incremental contribution of the paper . . . 3

**2 Data and measurement** **3**

2.1 Research setting: Fowa rug producers and export intermediary . . . 3

2.2 Experimental samples and treatment . . . 4

2.3 Production, profit, and productivity measures . . . 4

2.4 Quality measures . . . 5

2.5 Information-flow data . . . 5

**3 Model framework and empirical specifications** **5**

3.1 Firm performance and product specifications . . . 5

3.2 ITT specification . . . 5

3.3 TOT specification . . . 6

3.4 Five-step test for learning-by-exporting . . . 6

**4 Identification strategy** **6**

4.1 Randomization and balance . . . 6

4.2 Why detailed measurement matters . . . 6

4.3 Threats and how the paper addresses them . . . 7

**5 Tables and figures** **8**

5.1 Figures I–II: Export specifications and domestic versus export rugs . . . 8

5.2 Figure III and Table I: Cumulative export orders and sample/takeup statistics . . . 8

5.3 Tables II–III: Survey timeline and baseline balance . . . 9

5.4 Tables IV–V: Export participation and firm profits . . . 10

5.5 Tables VI–VII: Efficiency, scale, product mix, and inputs . . . 11

5.6 Tables VIII–IX: Quality levels and unadjusted productivity . . . 11

5.7 Figure IV and Table X: Specification overlap and specification-adjusted performance . . . 12

5.8 Table XI: Identical-specification quality lab . . . 13

5.9 Figure V and Table XII: Learning curves and information flows . . . 14

**6 Interpretive synthesis** **15**

**7 Short summary** **16**

**8 Conclusion** **16**

8.1 Core conclusion . . . . . 16
8.2 Economic intuition . . . . . 16
8.3 Incremental contribution in one sentence . . . . . 16
8.4 Limitations . . . . . 16
8.5 Final takeaway . . . . . 16

# 1 Big picture

- **Question.** Does exporting causally improve firm performance, and if performance rises, is it due to true learning-by-exporting or only movement along the existing production possibility frontier?
- **Setting.** The paper studies small handmade rug producers in Fowa, Egypt, and gives a randomly selected subset the opportunity to produce export orders for high-income markets through a local intermediary, Hamis Carpets.
- **Experiment.** The intervention reduces matching frictions between foreign buyers and small producers. Treatment firms are offered an initial export order, and if successful they can receive subsequent export orders.
- **Main performance result.** Treatment firms report roughly 16–26% higher profits relative to control firms.
- **Quality and output result.** Exporting induces large quality upgrading and higher output/input prices, but unadjusted productivity measures such as output per hour fall because export rugs are more difficult and time-consuming.
- **Learning result.** The paper finds evidence of genuine learning-by-exporting: conditional on product specifications, quality and productivity rise; treatment firms produce higher-quality identical domestic rugs in a quality lab without taking longer; learning curves appear over cumulative exports; and knowledge flows from buyers/intermediary predict quality improvements in the relevant dimensions.
- **Mechanism.** High-income foreign buyers demand higher-quality, modern-design rugs. The export relationship transfers technical knowledge about how to produce those rugs, partly through the intermediary.
- **Policy implication.** Export-promotion programs can generate productivity gains among small firms, not only demand shocks, but the mechanism depends on knowledge-intensive buyer relationships.

## 1.1 Incremental contribution of the paper

- **First randomized experiment on exporting.** Prior work struggled with selection because more productive firms choose to export. This paper creates exogenous firm-level variation in access to export markets through an RCT.
- **Separates product upgrading from learning.** Many studies use revenue productivity or coarse product codes. This paper collects product-line specifications, quality measures, and a quality-lab experiment to distinguish movement along the PPF from an outward shift of the PPF.
- **Micro mechanism of learning-by-exporting.** The paper does not just show that exporters perform better. It documents how learning occurs: through buyer/intermediary feedback, discussions about techniques, and dimension-specific quality improvements.
- **Firm-level development policy evaluation.** The paper evaluates an export-market-access intervention that resembles actual aid-for-trade programs and intermediary-based exporting used by small firms.
- **Rich measurement innovation.** The combination of repeated surveys, detailed rug specifications, direct quality scoring, identical-input quality-lab production, and correspondence/logbook data is the paper's distinctive empirical contribution.

# 2 Data and measurement

## 2.1 Research setting: Fowa rug producers and export intermediary

The study is conducted in Fowa, a peri-urban Egyptian town known for a cluster of small flat-weave rug producers. Most firms are very small, often consisting of a single owner working with family or hired labor

on wooden foot-treadle looms. The authors partner with Aid to Artisans and a local intermediary, Hamis Carpets, to find foreign buyers and transmit export orders to producers.

- Producers specialize in rug types such as duble, tups, kasaees, and goublan.
- Export buyers demand different designs and specifications from domestic buyers.
- The intermediary provides design-specific inputs such as thread, reed, and heddle, and communicates with firms about quality and technique.
- Exporting occurs indirectly through Hamis, which is realistic for small firms that cannot bear fixed costs of direct exporting.

**Interpretation:** The setting is valuable because producers use comparable technologies and because the researchers can observe highly detailed product specifications, allowing productivity and quality to be measured more carefully than in standard firm data.

## 2.2 Experimental samples and treatment

The experiment includes two main duble-rug samples plus smaller samples of other rug types. Treatment firms are offered an initial export order. Control firms are not contacted by Hamis for export orders during the experiment.

- **Sample 1.** The initial sample includes producers across rug types. Takeup is low for non-duble producers because foreign demand is concentrated in duble designs.
- **Sample 2.** After large sustained export orders arrive, the authors locate additional duble producers and randomize a second sample.
- **Takeup.** Treatment creates an opportunity to export, but not all treated firms take up the opportunity; this motivates ITT and TOT specifications.
- **Follow-up.** Firms are surveyed repeatedly at roughly four-month intervals. A quality lab is conducted at the end of the experiment.

**Interpretation:** The experiment mimics a real buyer-seller relationship rather than a one-shot subsidy. Subsequent export orders depend on firms' ability to satisfy the intermediary and buyers.

## 2.3 Production, profit, and productivity measures

The survey records production activity in the month before each interview. Variables include revenues, direct profits, expenses, output quantity, output prices, input prices and quantities, labor hours, rug type, and detailed rug specifications. Nominal variables are converted to real values using official Egyptian CPI.

The paper uses three productivity families:

- output per hour;
- TFP-style productivity measures based on observed inputs;
- specification-adjusted productivity, which controls for codified rug attributes.

**Interpretation:** Unadjusted productivity can fall when firms switch to more demanding rugs. Therefore, the paper's learning test relies on specification-adjusted productivity and on the quality-lab design.

### 2.4 Quality measures

Surveyors bring a master artisan to score rugs produced by firms on 11 quality dimensions. At the end of the study, all firms are invited to produce the same domestic rug using identical inputs and the same capital equipment in a leased workshop. Rugs are then assessed by both a master artisan and a professor of handicraft science.

- Survey quality metrics capture quality of rugs firms actually produce in normal production.
- Quality-lab metrics eliminate variation in product specifications, input quality, equipment, and scale.
- Objective measures include accuracy in length, width, and weight, and time to produce the rug.

**Interpretation:** The quality lab is the most direct way to ask whether treatment firms became more technically capable, rather than merely producing different products.

### 2.5 Information-flow data

The authors collect buyer-intermediary correspondence and a logbook of discussions between the intermediary and producers. They code whether the intermediary discussed particular quality dimensions, whether the discussion involved specific techniques, and whether mistakes were pointed out.

**Interpretation:** This information-flow data is unusual in the learning-by-exporting literature. It allows the paper to connect the content of knowledge transfers to dimension-specific quality improvements.

## 3 Model framework and empirical specifications

### 3.1 Firm performance and product specifications

The paper represents firm profits as:

$$\max_{l,k,\lambda} \pi = px - wl - F,$$

where output and quality are:

$$x = a(\lambda; \chi_a) f(l,k), \qquad q = q(\lambda; \chi_q), \qquad p = p_0 + bq(\lambda).$$

Here $\lambda$ indexes product specifications, $\chi_a$ is output efficiency, and $\chi_q$ is quality efficiency.

**Interpretation:** Exporting can raise profits by moving along the PPF: foreign buyers pay more for high-$\lambda$, high-quality rugs. Learning-by-exporting means exporting changes $\chi_a$ or $\chi_q$, shifting the PPF outward.

### 3.2 ITT specification

A simplified version of the ITT regression is:

$$y_{it} = \alpha_1 + \beta_1 Treatment_i + \gamma_1 y_{i0} + \delta_s + \tau_t + \varepsilon_{it},$$

where $Treatment_i$ is randomized assignment, $y_{i0}$ is the baseline value of the dependent variable, $\delta_s$ are randomization strata fixed effects, and $\tau_t$ are survey-round fixed effects.

**Interpretation:** The ITT estimates the effect of being offered the opportunity to export. It is policy-relevant because an export-promotion program can offer access but cannot force firms to take up the opportunity.

### 3.3 TOT specification

The treatment-on-treated specification uses takeup and instruments takeup with randomized treatment assignment:

$$y_{it} = \alpha_2 + \beta_2 Takeup_{it} + \gamma_2 y_{i0} + \delta_s + \tau_t + \nu_{it}.$$

**Interpretation:** Because takeup is endogenous, randomized treatment is used as an instrument. TOT effects are larger than ITT effects because they scale the impact to firms that actually produce export orders.

### 3.4 Five-step test for learning-by-exporting

The paper's learning test proceeds in five steps:

1. Control flexibly for rug specifications and test whether quality and productivity still rise.
2. Ask all firms to produce an identical domestic-market rug in the quality lab.
3. Estimate learning curves as quality/productivity residuals evolve with cumulative export production.
4. Link quality improvements to the dimensions discussed between intermediary and producer.
5. Rule out alternative explanations such as adjustment costs, scale effects, and unobserved investments.

**Interpretation:** These steps are designed to rule out the hypothesis that exporting only moves firms along the existing PPF. The evidence points to technical efficiency improvements, especially in quality production.

## 4 Identification strategy

### 4.1 Randomization and balance

The core identification comes from random assignment of export opportunities across firms. Baseline balance tests show that treatment and control firms are similar on household and firm characteristics. Survey attrition and quality-lab attrition are also similar across groups.

**Interpretation:** Randomization addresses the classic exporter-selection problem. More productive firms do not choose into the treatment group; the opportunity to export is assigned by the researchers.

### 4.2 Why detailed measurement matters

Even with randomization, the paper must distinguish true productivity improvements from compositional changes. Exporting changes the type of rugs produced, and export rugs are more difficult and higher quality. Without detailed specifications and quality measures, output per hour or revenue productivity could be misleading.

**Interpretation:** The experiment identifies the causal effect of export access, but measurement identifies the mechanism.

### 4.3 Threats and how the paper addresses them

- **Selective takeup:** handled with ITT and IV-based TOT.
- **Demand shock versus learning:** handled by specification controls, quality lab, learning curves, and information-flow evidence.
- **Specification measurement error:** addressed by the identical-rug quality lab.
- **Scale effects:** addressed by checking employees, looms, input quantities, and quality-lab production with common scale.
- **Investment explanations:** addressed by evidence that firms did not make large monetary or time investments in upgrading.
- **Intermediary selection of better firms for subsequent orders:** initial access is randomized; later selection may make treatment effects heterogeneous, but the ITT remains valid for the program.

**Interpretation:** The most convincing evidence is not any single estimate, but the consistency across random assignment, product-specification controls, lab production, learning curves, and observed knowledge transfers.

# 5 Tables and figures

## 5.1 Figures I–II: Export specifications and domestic versus export rugs

**Read:** Figure I shows the kind of detailed specifications transmitted from buyers to the intermediary and producers. Figure II shows that export rugs differ visibly from domestic rugs in design and style.

**Interpretation:** These figures motivate the measurement problem. Exporting changes the rug's attributes, so output per hour falls partly because producers make different, higher-specification products.

**Economic message:** The buyers are not just buying more rugs. They are buying a different quality/style bundle, which creates scope for both quality upgrading and learning.

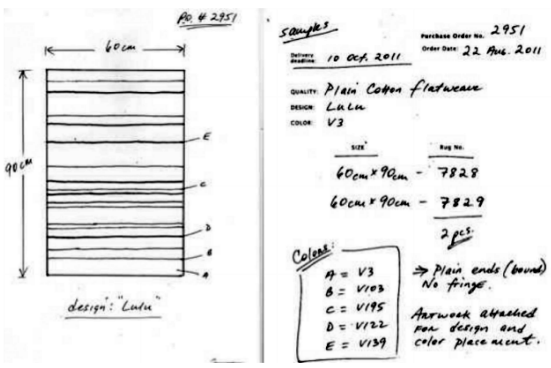


FIGURE I

Example of Rug Specifications Provided by a Potential Foreign Client

**(a)** Figure I: Example of rug specifications.

FIGURE II

Examples of Domestic Rugs and Export Rugs

Figure provides examples of the domestic rug (left) and export rugs ordered by foreign buyers (center, right). The domestic rug is the duble rug that firms were asked to manufacture at the quality lab.

**(b)** Figure II: Domestic and export rugs.

## 5.2 Figure III and Table I: Cumulative export orders and sample/takeup statistics

**Read:** Figure III shows cumulative export orders rising sharply after the first large order in June 2012. Table I describes the experimental samples and takeup. In the joint duble sample, 74 of 219 firms are treated and 46 take up; mean output conditional on takeup is 538 $m^2$.

**Interpretation:** The figure confirms that the export opportunity becomes economically meaningful. Table I shows that takeup is strong for duble firms, but lower for other rug types because foreign demand is concentrated in export-style duble rugs.

**Economic message:** The treatment is not just a small information nudge. It generates sustained export production for a subset of treated firms.

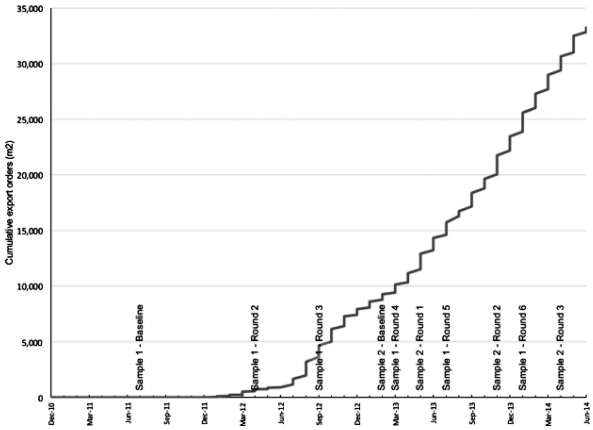


FIGURE III

**Cumulative Export Orders**

**(a)** Figure III: Cumulative export orders.

TABLE I

SAMPLE AND TAKEUP STATISTICS

| | Duble orders | | | | | Kasaees orders |
|---|---|---|---|---|---|---|
| | Joint sample | Sample 1 | | | Sample 2 | Sample 1 |
| Statistic | Duble firms (1) | Goublan firms (2) | Tups firms (3) | Duble firms (4) | Duble firms (5) | Kasaees firms (6) |
| Firms | 219 | 103 | 83 | 79 | 140 | 38 |
| Treatment firms | 74 | 49 | 42 | 39 | 35 | 19 |
| Takeup firms | 46 | 5 | 8 | 14 | 32 | 5 |
| Initial packet size ($m^2$) | 110 | 110 | 110 | 110 | 110 | 250 |
| Successful takeup firms | 46 | 4 | 6 | 14 | 32 | 5 |
| Mean output conditional on takeup ($m^2$) | 538 | 586 | 589 | 778 | 434 | 303 |
| Std. dev. of output conditional on takeup | 188 | 174 | 208 | 132 | 177 | 76 |

*Notes.* Table reports statistics by firm type and sample. The first row displays the number of firms within each rug type and sample. The second row displays the number of firms in the treatment group. The third row indicates the number of firms who accepted the treatment and agreed to make rugs for export. The fourth row is the initial order size (in square meters) offered to each takeup firm. The fifth row shows the number of firms that completed the initial order successfully and received subsequent orders from Hamis. The sixth row indicates average output conditional on takeup. The last row reports the standard deviation of output in the first year, conditional on takeup.

**(b)** Table I: Sample and takeup statistics.

## 5.3 Tables II–III: Survey timeline and baseline balance

**Read:** Table II lays out the repeated survey schedule and the timing of the quality lab. Table III shows treatment and control firms are balanced at baseline: the joint $F$-test is 1.23 and attrition is similar across groups.

**Interpretation:** Table II establishes that the data are panel data with repeated follow-ups, not a single endline survey. Table III supports the validity of the randomization.

**Economic message:** The identification is credible because treatment does not appear to predict baseline firm quality, profits, household characteristics, or attrition.

TABLE II

Survey Timeline

| Survey timeline | Sample 1 | Sample 2 |
|---|---|---|
| Baseline round 0 | July–Aug 2011[a] | Feb–Mar 2013[a] |
| Follow-up round 1 | Nov–Dec 2011[b] | May–June 2013 |
| Follow-up round 2 | April–May 2012 | Nov–Dec 2013 |
| Follow-up round 3 | Sept–Dec 2012 | May–June 2014[a] |
| Follow-up round 4 | Mar–Apr 2013 | |
| Follow-up round 5 | July–Oct 2013[a] | |
| Follow-up round 6 | Jan–Mar 2014 | |
| Quality lab | June 2014 | June 2014 |

*Notes.* Table reports the timeline for the data collection on duble firms by sample.

[a]Supplementary questions about household and firm outcomes are included in both samples' baseline round 0 surveys, in follow-up round 5 for Sample 1, and follow-up round 3 for Sample 2.

[b]The Egyptian survey company we hired for the Sample 1 baseline trained a new enumerator for the first Sample 1 follow-up. Unfortunately, we discovered that this enumerator did not actually interview all of the firms and entered in fake data for some, so we discard the data from follow-up round 1 for Sample 1. We immediately fired the enumerator and hired new employees in January 2012 who conducted all subsequent surveys. We managed these employees directly and implemented a stricter auditing procedure, as well as back checks using external evaluators, to ensure data integrity.

**(a)** Table II: Survey timeline.

TABLE III

Baseline Balance

| | Control group mean | Difference in treatment | $N$ |
|---|---|---|---|
| Panel A: Household characteristics | | | |
| Age | 51.0 | 0.9 | 218 |
| | (0.7) | (1.6) | |
| Number of years in rug business | 37.7 | 0.2 | 213 |
| | (0.8) | (1.7) | |
| Illiterate? | 0.63 | 0.10 | 214 |
| | (0.03) | (0.07) | |
| Household size | 4.2 | 0.0 | 219 |
| | (0.1) | (0.2) | |
| Household income | 1090.0 | 76.5 | 219 |
| | (91.2) | (228.0) | |
| Digit span recall | 5.8 | 0.2 | 204 |
| | (0.1) | (0.2) | |
| Panel B: Firm characteristics | | | |
| Price per square meter | 30.2 | 6.8 | 218 |
| | (3.3) | (7.8) | |
| Direct monthly profits from rug business | 646 | 7.9 | 218 |
| | (41.8) | (81.5) | |
| Reported monthly profits from rug business | 806 | −10.4 | 217 |
| | (38.5) | (84.4) | |
| Hours worked last month | 247 | −1.7 | 218 |
| | (5.6) | (11.7) | |
| Number of employees | 1.09 | 0.0 | 218 |
| | (0.0) | (0.1) | |
| Total produced last month ($m^2$) | 50.0 | 3.3 | 218 |
| | (4.3) | (10.0) | |
| Ever exported? | 0.12 | 0.02 | 219 |
| | (0.02) | (0.05) | |
| Average quality | 2.63 | −0.13*** | 218 |
| | (0.03) | (0.05) | |
| Joint $F$-test | 1.23 | | |
| Attrition in follow-up surveys | 0.11 | 0.00 | 815 |
| | (0.01) | (0.02) | |
| Attrition in quality lab | 0.14 | 0.02 | 219 |
| | (0.03) | (0.05) | |

*Notes.* Table explores baseline balance. Each row is a regression of the named variable on a constant, treatment dummy and strata fixed effects; the constant (control group mean) and treatment dummy are reported. The third-to-last row reports the $F$-test for a regression of the treatment dummy on all 14 baseline balance variables. Profits and prices are winsorized at the 1st and 99th percentile to trim outliers (without winsorizing, the sample still remains statistically balanced between treatment and control groups). The final rows report average attrition rates across all survey rounds and in the quality lab respectively. Significance:

**(b)** Table III: Baseline balance.

## 5.4 Tables IV–V: Export participation and firm profits

**Read:** Table IV shows the intervention increases the probability of knowingly exporting by 55 percentage points in the ITT and 76 percentage points in the TOT. Table V shows positive profit effects across direct, reported, constructed, and hypothetical profits.

**Interpretation:** Table IV is the first-stage/take-up result. Table V is the headline performance result: export access raises profits even though unadjusted output falls later.

**Economic message:** Access to high-income markets increases firm profitability, but the source of that gain is not simple quantity expansion.

TABLE IV

Impact of Intervention on Firms Knowingly Exporting

| | (1) ITT | (2) TOT |
|---|---|---|
| Indicator for ever exported | 0.55*** | 0.76*** |
| | (0.06) | (0.07) |
| $R$-squared | 0.33 | 0.45 |
| Control mean | 0.20 | 0.20 |
| Observations | 191 | 191 |

*Notes.* Table regresses an indicator for if a firm has ever knowingly produced rugs for export markets on indicators for treatment (column (1)) or takeup (column (2)). The question was asked in round 5 for Sample 1 and round 3 for Sample 2. The TOT regression instruments takeup with treatment. The regressions control for baseline values of the dependent variable, and include round and strata fixed effects. Significance: *.10; **.05; ***.01.

**(a)** Table IV: Firms knowingly exporting.

TABLE V

Impact of Exporting on Firm Profits

| | Log direct profits | | Log (reported revenues − reported costs) | | Log (constructed revenues − constructed costs) | | Log hypothetical profits | |
|---|---|---|---|---|---|---|---|---|
| | (1) ITT | (2) TOT | (3) ITT | (4) TOT | (5) ITT | (6) TOT | (7) ITT | (8) TOT |
| Panel A: Profits (in month prior to survey) | | | | | | | | |
| Treatment | 0.26*** | 0.42*** | 0.21*** | 0.37*** | 0.19*** | 0.34*** | 0.37*** | 0.68*** |
| | (0.05) | (0.08) | (0.06) | (0.10) | (0.06) | (0.10) | (0.11) | (0.19) |
| $R$-squared | 0.21 | 0.22 | 0.16 | 0.18 | 0.16 | 0.18 | 0.19 | 0.19 |
| Control mean (in levels) | 929 | 929 | 931 | 931 | 951 | 951 | 541 | 541 |
| Observations | 573 | 573 | 644 | 644 | 685 | 685 | 687 | 687 |
| Panel B: Profits per owner hour (in month prior to survey) | | | | | | | | |
| Treatment | 0.20*** | 0.32*** | 0.17*** | 0.29*** | 0.16*** | 0.28*** | 0.25*** | 0.46*** |
| | (0.05) | (0.08) | (0.05) | (0.09) | (0.05) | (0.09) | (0.07) | (0.12) |
| $R$-squared | 0.14 | 0.14 | 0.12 | 0.13 | 0.13 | 0.13 | 0.19 | 0.18 |
| Control mean (in levels) | 3.53 | 3.53 | 3.54 | 3.54 | 3.55 | 3.55 | 5.56 | 5.56 |
| Observations | 573 | 573 | 637 | 637 | 684 | 684 | 687 | 687 |

*Notes.* Table reports treatment effects on different measures of real profits in the month prior to the date of the survey, all measured in logs. See text for descriptions of each measure. Dependent variable in Panel A is profits. Dependent variable in Panel B is profits per owner hour. Owner hours include the hours of family member production when recorded. The regressions control for baseline values of the dependent variable, and include round and strata fixed effects. Control group means are reported in levels in Egyptian pounds (LE) in Panel A and LE/hour in Panel B. The TOT regressions instrument takeup with treatment. Standard errors are clustered by firm. Significance: *.10; **.05; ***.01.

**(b)** Table V: Firm profits.

## 5.5 Tables VI–VII: Efficiency, scale, product mix, and inputs

**Read:** Table VI decomposes output, output prices, labor hours, product variety, quality, and market segment. Table VII shows export-treated firms face higher input prices and changes in input use.

**Interpretation:** These tables show why simple productivity measures are misleading. Treatment firms produce fewer square meters, charge higher prices, use more labor hours, and move toward higher-quality market segments.

**Economic message:** Exporting induces quality upgrading, not just more output. The production bundle changes in a way that standard quantity productivity measures would misread.

TABLE VI

Sources of Changes to Firm Profits: Components of Profits

| | Log output price $\left(\frac{\text{LE}}{\text{m}^2}\right)$ | | Log output ($\text{m}^2$) | | Log hours worked | | Number of employees | | Log number of looms | | Log warp thread ball (kg) | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | (1) ITT | (2) TOT | (3) ITT | (4) TOT | (5) ITT | (6) TOT | (7) ITT | (8) TOT | (9) ITT | (10) TOT | (11) ITT | (12) TOT |
| Treatment | 0.43*** | 0.78*** | −0.26*** | −0.47*** | 0.05** | 0.08** | 0.01 | 0.01 | −0.02 | −0.04 | 0.15*** | 0.25*** |
| | (0.10) | (0.19) | (0.09) | (0.17) | (0.02) | (0.04) | (0.01) | (0.01) | (0.04) | (0.06) | (0.05) | (0.08) |
| $R$-squared | 0.16 | 0.15 | 0.24 | 0.22 | 0.12 | 0.13 | 0.02 | 0.02 | 0.13 | 0.13 | 0.24 | 0.24 |
| Control mean (in levels) | 28.2 | 28.2 | 64.1 | 64.1 | 269.0 | 269.0 | 1.0 | 1.0 | 1.1 | 1.1 | 6.0 | 6.0 |
| Observations | 691 | 691 | 676 | 676 | 678 | 678 | 695 | 695 | 694 | 694 | 600 | 600 |

*Notes.* Table reports treatment effects on output prices and quantities, hours, number of employees (inclusive of owner), looms, and the size of the warp thread ball (which is a proxy for the length of the production run), all measured in logs except number of employees. The TOT regressions instrument takeup with treatment. Hours worked are calculated using average daily hours and number of days worked last month. Control group means are reported in levels. The regressions control for baseline values of the dependent variable, and include round and strata fixed effects. Standard errors are clustered by firm. Significance: *.10; **.05; ***.01.

**(a)** Table VI: Efficiency, scale, and product mix.

TABLE VII

Sources of Changes to Firm Profits: Inputs

| | Log weft thread price $\left(\frac{\text{LE}}{\text{m}^2}\right)$ | | Log warp thread price $\left(\frac{\text{LE}}{\text{m}^2}\right)$ | | Log weft thread quantity (g) | | Log warp thread quantity (g) | |
|---|---|---|---|---|---|---|---|---|
| | (1) ITT | (2) TOT | (3) ITT | (4) TOT | (5) ITT | (6) TOT | (7) ITT | (8) TOT |
| Treatment | 0.20*** | 0.33*** | −0.04 | −0.07 | −0.19** | −0.34** | 0.04 | 0.06 |
| | (0.06) | (0.10) | (0.03) | (0.06) | (0.10) | (0.17) | (0.11) | (0.20) |
| $R$-squared | 0.22 | 0.24 | 0.27 | 0.27 | 0.23 | 0.22 | 0.10 | 0.11 |
| Control mean (in levels) | 12.8 | 12.8 | 18.1 | 18.1 | 110.0 | 110.0 | 17.8 | 17.8 |
| Observations | 564 | 564 | 685 | 685 | 677 | 677 | 686 | 686 |

*Notes.* Table analyzes input prices and quantities, all measured in logs. The TOT regressions instrument takeup with treatment. Control group means are reported in levels. The regressions control for baseline values of the dependent variable, and include round and strata fixed effects. Standard errors are clustered by firm. Significance: *.10; **.05; ***.01.

**(b)** Table VII: Input prices and quantities.

## 5.6 Tables VIII–IX: Quality levels and unadjusted productivity

**Read:** Table VIII shows large positive quality effects across almost every dimension. The stacked quality metric rises by 0.79 in the ITT and 1.35 in the TOT. Table IX shows unadjusted output per hour and TFP fall.

**Interpretation:** The tables should be read together: quality increases while unadjusted productivity falls because export products are more difficult and take longer to produce.

**Economic message:** If researchers only saw output per hour, they might conclude that exporting hurts productivity. Detailed quality data reveal that firms are moving toward more valuable products.

TABLE VIII

IMPACT OF EXPORTING ON QUALITY LEVELS

| | Control mean | (1) ITT | (2) TOT |
|---|---|---|---|
| Panel A: Quality metrics | | | |
| Corners | 2.98 | 1.11*** | 1.70*** |
| | | (0.12) | (0.11) |
| Waviness | 2.99 | 1.10*** | 1.68*** |
| | | (0.12) | (0.10) |
| Weight | 3.08 | 1.07*** | 1.63*** |
| | | (0.11) | (0.11) |
| Touch | 3.12 | 0.40*** | 0.66*** |
| | | (0.06) | (0.07) |
| Packedness | 3.11 | 0.89*** | 1.59*** |
| | | (0.11) | (0.12) |
| Warp thread tightness | 3.05 | 0.83*** | 1.49*** |
| | | (0.10) | (0.12) |
| Firmness | 2.98 | 0.87*** | 1.60*** |
| | | (0.11) | (0.12) |
| Design accuracy | 3.17 | 0.79*** | 1.41*** |
| | | (0.10) | (0.12) |
| Warp thread packedness | 3.05 | 1.07*** | 1.65*** |
| | | (0.11) | (0.11) |
| Inputs | 3.07 | 0.89*** | 1.62*** |
| | | (0.10) | (0.12) |
| Loom | 2.02 | 0.03 | 0.05 |
| | | (0.02) | (0.04) |
| $R$-squared | | 0.44 | 0.60 |
| Observations | | 6,885 | 6,885 |
| Panel B: Stacked quality metrics | | | |
| Stacked quality metrics | 2.96 | 0.79*** | 1.35*** |
| | | (0.09) | (0.08) |
| $R$-squared | | 0.39 | 0.54 |
| Observations | | 6,885 | 6,885 |

*Notes.* Panel A stacks the quality metrics and interacts treatment (ITT) or takeup (TOT) with a quality-metric indicator variable. The coefficients on the interactions provide the treatment effects separately for each metric. The TOT instruments takeup interacted with quality metric with treatment interacted with quality metric. Each regression includes baseline values of the quality metric, strata and round fixed effects, and each of these controls interacted with quality-metric. Panel B constrains the treatment effects to be equal across quality metrics; these regressions include baseline values, strata and round fixed effects. Control group means are reported in levels. Standard errors are clustered by firm. Significance: *.10; **.05; ***.01.

**(a)** Table VIII: Quality levels.

TABLE IX

IMPACT OF EXPORTING ON UNADJUSTED PRODUCTIVITY

| | Log unadjusted output per hour | | Log unadjusted TFP | |
|---|---|---|---|---|
| | (1) ITT | (2) TOT | (3) ITT | (4) TOT |
| Treatment | −0.24*** | −0.42*** | −0.28*** | −0.50*** |
| | (0.09) | (0.16) | (0.09) | (0.16) |
| $R$-squared | 0.18 | 0.16 | 0.26 | 0.24 |
| Control mean (in levels) | 0.26 | 0.26 | 0.49 | 0.49 |
| Observations | 687 | 687 | 674 | 674 |

*Notes.* Table reports treatment effects for the two productivity measures: log unadjusted output per labor hour (in $\frac{m^2}{hour}$) and log unadjusted TFP. See text and Appendix for the methodology used to obtain unadjusted TFP. The TOT specifications instrument takeup with treatment. Control group means are reported in levels. Regressions control for baseline values of the variable, round and strata fixed effects. Standard errors are clustered by firm. Significance: *.10; **.05; ***.01.

quality-upgrading mechanisms, and the distinction is important for understanding how exporting improves firm performance. In the first mechanism, firms always knew how to manufacture the high-quality rugs demanded by rich-country buyers. If foreign buyers pay higher prices, but particularly so for high-quality prod-

**(b)** Table IX: Unadjusted productivity.

## 5.7 Figure IV and Table X: Specification overlap and specification-adjusted performance

**Read:** Figure IV shows overlap in rug specifications between domestic and export orders, which makes it possible to control for specifications. Table X shows that after controlling for specifications, both quality and productivity improve.

**Interpretation:** This is the first step in detecting learning-by-exporting. If exporting only moved firms along the PPF, specification-adjusted productivity and quality would not rise.

**Economic message:** Conditional on producing comparable products, exporters become technically more efficient.

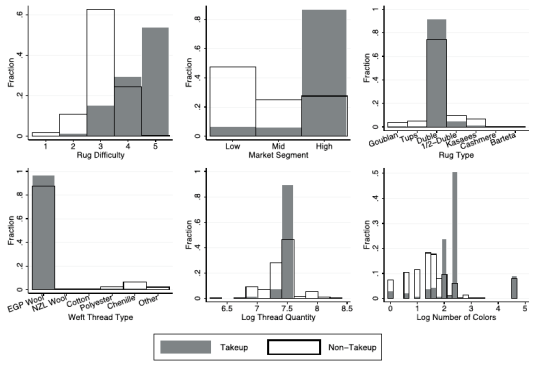


FIGURE IV

Overlap in Rug Specifications between Domestic and Export Orders

s the density of the six rug specifications for takeup (shaded) and nontakeup (outline) firms.

**(a)** Figure IV: Specification overlap.

TABLE X

QUALITY AND PRODUCTIVITY CONDITIONAL ON SPECIFICATIONS (STEP 1)

| | Stacked quality metrics | | Log output per hour | | Log TFP | |
|---|---|---|---|---|---|---|
| | (1) ITT | (2) TOT | (3) ITT | (4) TOT | (5) ITT | (6) TOT |
| Panel A: Specification controls | | | | | | |
| Treatment | 0.32*** | 0.78*** | 0.18** | 0.44** | 0.14** | 0.35** |
| | (0.04) | (0.08) | (0.08) | (0.18) | (0.07) | (0.16) |
| (log) Thread quantity | 0.04 | 0.02 | −0.12 | −0.13 | −0.07 | −0.08 |
| | (0.05) | (0.04) | (0.13) | (0.13) | (0.13) | (0.12) |
| Difficulty control | 0.47*** | 0.34*** | −0.14*** | −0.21*** | −0.16*** | −0.22*** |
| | (0.02) | (0.03) | (0.04) | (0.05) | (0.04) | (0.05) |
| (log) # colors | 0.03** | 0.01 | −0.05* | −0.07** | −0.06** | −0.07*** |
| | (0.01) | (0.01) | (0.03) | (0.03) | (0.03) | (0.02) |
| Low-market segment | −0.19*** | −0.08** | 0.43*** | 0.49*** | 0.42*** | 0.47*** |
| | (0.03) | (0.03) | (0.08) | (0.09) | (0.07) | (0.08) |
| Mid-market segment | −0.19*** | −0.05 | 0.29*** | 0.36*** | 0.26*** | 0.32*** |
| | (0.04) | (0.04) | (0.08) | (0.09) | (0.07) | (0.08) |
| Rug type FEs | Yes | Yes | Yes | Yes | Yes | Yes |
| Input thread type FEs | Yes | Yes | Yes | Yes | Yes | Yes |
| $R$-squared | 0.64 | 0.67 | 0.57 | 0.58 | 0.62 | 0.63 |
| Observations | 6,820 | 6,820 | 673 | 673 | 660 | 660 |
| Panel B: Specification fixed effects | | | | | | |
| Treatment | 0.13** | 0.53*** | 0.31*** | 1.26*** | 0.25*** | 0.94*** |
| | (0.05) | 0.17 | (0.08) | (0.45) | (0.08) | (0.36) |
| Specification FEs | Yes | Yes | Yes | yes | Yes | Yes |
| $R$-squared | 0.71 | 0.72 | 0.69 | 0.62 | 0.70 | 0.68 |
| Observations | 6,820 | 6,820 | 428 | 428 | 416 | 416 |
| Panel C: Specification-adjusted dependent variables | | | | | | |
| Treatment | 0.42*** | 0.72*** | 0.18** | 0.33*** | 0.20*** | 0.36*** |
| | (0.05) | (0.04) | (0.07) | (0.13) | (0.07) | (0.12) |
| $R$-squared | 0.18 | 0.27 | 0.06 | 0.10 | 0.13 | 0.18 |
| Observations | 6,860 | 6,860 | 678 | 678 | 669 | 669 |

*Notes.* Table reports treatment effects for the stacked quality measures and the two productivity measures after including various controls for the specifications of the rug on the loom at the time of the survey. The TOT specifications instrument takeup with treatment. In Panel A, there are both rug and input thread type fixed effects in addition to the specification controls included in the table. Panel B uses fixed effects for each of the 435 combinations of the six specification controls. Both panels use unadjusted productivity measures. Panel C uses specification-adjusted measures for quality and productivity; see text and Appendix for details. Regressions in all panels also control for baseline values of the dependent variable (and baseline values of the specification controls in Panel A), as well as round and strata fixed effects. Standard errors are clustered by firm. Significance: *.10; **.05; ***.01.

**(b)** Table X: Specification-adjusted quality and productivity.

## 5.8 Table XI: Identical-specification quality lab

**Read:** Table XI asks all firms to produce the same domestic rug using identical inputs and a common loom. Treatment firms produce higher-quality rugs according to both expert assessors and objective accuracy measures. They do not take longer to produce the rug.

**Interpretation:** This is the cleanest mechanism test. Product specifications, inputs, equipment, and scale are held fixed; the remaining difference is technical capability.

**Economic message:** Exporting appears to teach firms how to produce quality, not merely to choose different product varieties.

TABLE XI

QUALITY AND PRODUCTIVITY ON IDENTICAL-SPECIFICATION DOMESTIC RUGS (STEP 2)

| | Master artisan | | | Professor | | |
|---|---|---|---|---|---|---|
| | Control mean | (1) ITT | (2) TOT | Control mean | (3) ITT | (4) TOT |
| Panel A: Quality metrics | | | | | | |
| Corners | 3.23 | 0.72*** | 1.05*** | 3.31 | 0.29** | 0.43** |
| | | (0.14) | (0.17) | | (0.13) | (0.18) |
| Waviness | 3.17 | 0.55*** | 0.80*** | 3.31 | 0.25** | 0.36** |
| | | (0.14) | (0.18) | | (0.12) | (0.16) |
| Weight | 3.60 | 0.62*** | 0.91*** | 3.64 | 0.58*** | 0.86*** |
| | | (0.13) | (0.16) | | (0.17) | (0.25) |
| Packedness | 3.30 | 0.77*** | 1.14*** | 3.28 | 0.28** | 0.42*** |
| | | (0.13) | (0.15) | | (0.11) | (0.15) |
| Touch | 3.29 | 0.52*** | 0.76*** | 3.27 | 0.36*** | 0.52*** |
| | | (0.11) | (0.14) | | (0.12) | (0.16) |
| Warp thread tightness | 3.00 | 0.51*** | 0.74*** | 3.30 | 0.25** | 0.36** |
| | | (0.09) | (0.11) | | (0.12) | (0.16) |
| Firmness | 3.21 | 0.71*** | 1.04*** | 3.23 | 0.29** | 0.43*** |
| | | (0.14) | (0.17) | | (0.12) | (0.16) |
| Design accuracy | 3.65 | 0.53*** | 0.77*** | 3.45 | 0.27** | 0.40*** |
| | | (0.11) | (0.15) | | (0.11) | (0.15) |
| Warp thread packedness | 3.05 | 0.87*** | 1.28*** | 3.20 | 0.39*** | 0.58*** |
| | | (0.14) | (0.17) | | (0.12) | (0.16) |
| $R$-squared | | 0.21 | 0.34 | | 0.11 | 0.14 |
| Observations | | 1,680 | 1,680 | | 1,667 | 1,667 |
| Panel B: Stacked quality metrics | | | | | | |
| Stacked quality metric | 3.28 | 0.64*** | 0.94*** | 3.33 | 0.33*** | 0.48*** |
| | | (0.10) | (0.12) | | (0.10) | (0.13) |
| $R$-squared | | 0.19 | 0.32 | | 0.09 | 0.13 |
| Observations | | 1,680 | 1,680 | | 1,667 | 1,667 |
| Panel C: Objective metrics | | | | | | |
| | Control mean | (1) ITT | (2) TOT | | | |
| Length accuracy | −4.51 | 1.43*** | 2.09*** | | | |
| | | (0.51) | (0.71) | | | |
| Width accuracy | −2.29 | 0.17 | 0.25 | | | |
| | | (0.29) | (0.41) | | | |
| Weight accuracy | −221.0 | 89.1*** | 131.0*** | | | |
| | | (20.3) | (29.6) | | | |

**(a)** Table XI: quality-lab results, part 1.

TABLE XI

(CONTINUED)

| | Control mean | (1) ITT | (2) TOT |
|---|---|---|---|
| Time (in minutes) | 247.0 | −5.67 | −8.3 |
| | | (6.6) | (9.5) |
| $R$-squared | | 0.84 | 0.84 |
| Observations | | 748 | 748 |

*Notes.* Table reports ITT and TOT specifications using the nine quality metrics from the quality lab. Panel A stacks the quality metrics and interacts treatment (ITT) or takeup (TOT) with a quality-metric indicator variable. The coefficients on the interactions provide the treatment effects separately for each metric. The TOT instruments takeup (interacted with quality metric) with treatment (also interacted with quality metric). Panel B constrains the treatment effects to be equal across quality metrics. Columns (1) and (2) report scores from the master artisan. Columns (3) and (4) report scores from the professor of Handicraft Science at Domietta University. Panel C reports objective accuracy measures, which are calculated as the negative of the absolute error for that specification, so that a higher value indicates that the manufactured rug was closer to intended length (140 cm), width (70 cm), and weight (1,750 g). It also includes the time spent to produce the rug in minutes. As in Panel A, these are run in a single regression by intereacting the objective measure with treatment or takeup. All regressions include interactions of strata fixed effects with quality-metric indicators, and standard errors are clustered by firm. Significance: *.10; **.05; ***.01.

**(b)** Table XI: quality-lab results, part 2.

## 5.9 Figure V and Table XII: Learning curves and information flows

**Read:** Figure V plots learning curves over cumulative exports among takeup firms. Quality improves with cumulative production, and specification-adjusted productivity also rises. Table XII shows that quality improves more along dimensions that the intermediary discussed with producers, especially when discussions involved specific techniques or pointed-out mistakes.

**Interpretation:** These are the direct learning mechanisms. Figure V shows gradual accumulation; Table XII links improvements to the actual content of knowledge transferred through buyer/intermediary-producer interactions.

**Economic message:** Learning-by-exporting is not a black box: high-income buyers and intermediaries transmit production knowledge that improves firm capabilities.

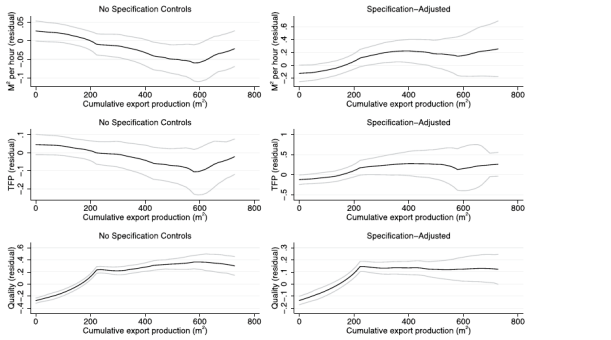


FIGURE V

Learning Curves for Takeup Firms (Step 3)

learning curves obtained by regressing quality or productivity measures on firm and round fixed effects and then plotting ted local polynomial of the residuals against cumulative export production for takeup firms. Figure restricts attention to mulative exports achieved by Sample 2 firms.

**(a)** Figure V: Learning curves for takeup firms.

TABLE XII

INFORMATION FLOWS AND QUALITY LEVELS (STEP 4)

| | (1) Stacked quality metrics | (2) Specification-adjusted quality metrics | (3) Stacked quality metrics | (4) Specification-adjusted quality metrics |
|---|---|---|---|---|
| Panel A: Baseline | | | | |
| *Takeup*$_i$ × *Talked About Dimension*$_{id}$ | 0.19** | 0.16** | | |
| | (0.08) | (0.07) | | |
| *Takeup*$_i$ × *Information on Techniques for Dimension*$_{id}$ | | | 0.32*** | 0.33*** |
| | | | (0.09) | (0.09) |
| *Takeup*$_i$ × *Pointed Out Mistakes in Dimension*$_{id}$ | | | 0.30*** | 0.29*** |
| | | | (0.04) | (0.04) |
| Quality metric FEs | Yes | Yes | Yes | Yes |
| *Takeup*$_i$ × quality metric FEs | No | No | No | No |
| Firm FEs | Yes | Yes | Yes | Yes |
| Specification-adjusted quality metrics | No | Yes | No | Yes |
| $R$-squared | 0.76 | 0.43 | 0.75 | 0.42 |
| Observations | 1,700 | 1,667 | 1,670 | 1,637 |

**(b)** Table XII: information flows and quality, part 1.

TABLE XII

CONTINUED

| | (1) Stacked quality metrics | (2) Specification-adjusted quality metrics | (3) Stacked quality metrics | (4) Specification-adjusted quality metrics |
|---|---|---|---|---|
| Panel B: With "takeup × quality" FEs | | | | |
| *Takeup*$_i$ × *Talked About Dimension*$_{id}$ | 0.15*** | 0.13** | | |
| | (0.05) | (0.05) | | |
| *Takeup*$_i$ × *Information on Techniques for Dimension*$_{id}$ | | | 0.16* | 0.16 |
| | | | (0.09) | (0.10) |
| *Takeup*$_i$ × *Pointed Out Mistakes in Dimension*$_{id}$ | | | 0.17** | 0.15* |
| | | | (0.08) | (0.08) |
| Quality metric FEs | Yes | Yes | Yes | Yes |
| *Takeup*$_i$ × quality metric FEs | Yes | Yes | Yes | Yes |
| Firm FEs | Yes | Yes | Yes | Yes |
| Specification-adjusted quality metrics | No | Yes | No | Yes |
| $R$-squared | 0.81 | 0.56 | 0.81 | 0.55 |
| Observations | 1,700 | 1,667 | 1,670 | 1,637 |

*Notes.* Table regresses stacked quality metrics on takeup indicator and its interaction with a dummy that takes the value 1 if the intermediary talked to the firm about that particular dimension of quality. Even-numbered columns use the specification-adjusted quality metrics described in the text. Columns (3) and (4) separate whether the discussion along that dimension was about technique or just pointing out mistakes. Regressions are run on a cross-section of firms and include baseline values, firm fixed effects, quality metric fixed effects, and quality metric fixed effects interacted with takeup. Panel B repeats the regressions in Panel A but also includes takeup interacted with quality metric fixed effects. Standard errors are clustered by firm. Significance: *.10; **.05; ***.01.

Table XII: information flows and quality, part 2.

# 6 Interpretive synthesis

- **First layer: demand effect.** The export opportunity raises demand for high-quality rugs and therefore raises profits.
- **Second layer: product-upgrading effect.** Firms change what they make: higher-quality, more difficult, more input-intensive rugs.
- **Third layer: learning effect.** Firms improve how they make these rugs. The evidence comes from specification controls, quality-lab output, learning curves, and information-flow matches.
- **Why this matters for trade theory.** If exporting improves capabilities, gains from trade are larger than in static models where firms only reallocate along a fixed PPF.
- **Why this matters for policy.** Export promotion can be productivity policy if it creates sustained relationships with knowledgeable buyers, not merely access to a one-time order.

# 7 Short summary

- The paper conducts a randomized experiment that gives Egyptian rug producers export-market access.
- Treatment raises the probability of knowingly exporting and increases profits.
- Exporting induces quality upgrading and higher prices, but unadjusted output per hour falls.
- Detailed product specifications and the quality lab distinguish quality upgrading from technical learning.
- Specification-adjusted quality and productivity increase.
- In an identical domestic-rug lab, treatment firms produce higher-quality rugs without taking longer.
- Learning curves show gradual improvement with cumulative export production.
- Information-flow data show that quality improves most along the dimensions discussed with the intermediary.
- The paper concludes that learning-by-exporting exists in this setting and operates partly through knowledge transfer from buyers and intermediaries.

# 8 Conclusion

## 8.1 Core conclusion

Atkin, Khandelwal, and Osman provide experimental evidence that exporting causally improves firm performance and induces learning-by-exporting among small Egyptian rug producers. Exporting raises profits and quality, and the learning evidence survives specification controls, an identical-input quality lab, learning-curve analysis, and information-flow tests.

## 8.2 Economic intuition

Foreign buyers demand products that are different from and more demanding than domestic products. This creates incentives to upgrade quality and exposes firms to technical knowledge. Exporting therefore changes both what firms produce and what they know how to produce.

## 8.3 Incremental contribution in one sentence

The paper moves the learning-by-exporting literature from observational correlations to an RCT with direct product-specification, quality, productivity, and knowledge-transfer evidence.

## 8.4 Limitations

The experiment studies small handmade producers in one Egyptian cluster, exporting indirectly through an intermediary. External validity to large automated manufacturers or direct exporters requires caution. The treatment combines access to foreign demand with buyer/intermediary feedback, so the paper identifies the effect of a realistic export opportunity rather than a pure price or pure information shock.

## 8.5 Final takeaway

Export market access can be a capability-building intervention when it connects small firms to sophisticated buyers whose demands and feedback transmit production knowledge.